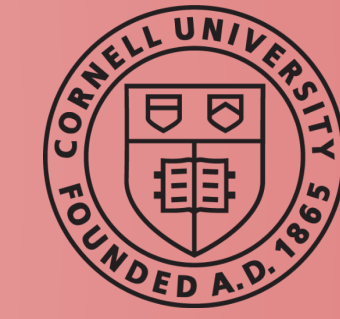


Stochastic Depth in ResNets

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Problem: Deep networks are good but hard to train.

Do we need all layers during training?

Residual Networks

$$H_l = \text{ReLU}(f(H_{l-1}) + \text{id}(H_{l-1}))$$

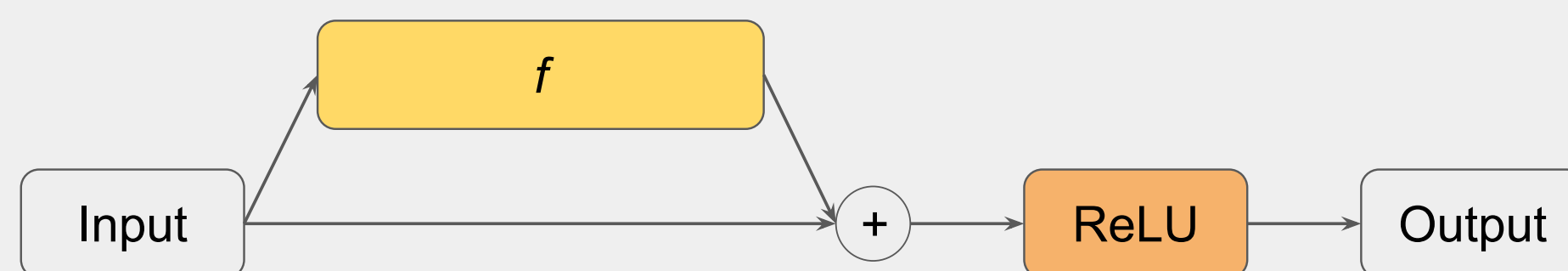


Figure 1. Residual block with identity skip connection.

Stochastic Depth

$$b_l \sim \text{Bernoulli}(p_l)$$

$$H_l = \text{ReLU}(b_l f(H_{l-1}) + \text{id}(H_{l-1}))$$

$$p_l = \Pr(b_l = 1) \\ = 1 - (l/L)(1 - p_L)$$

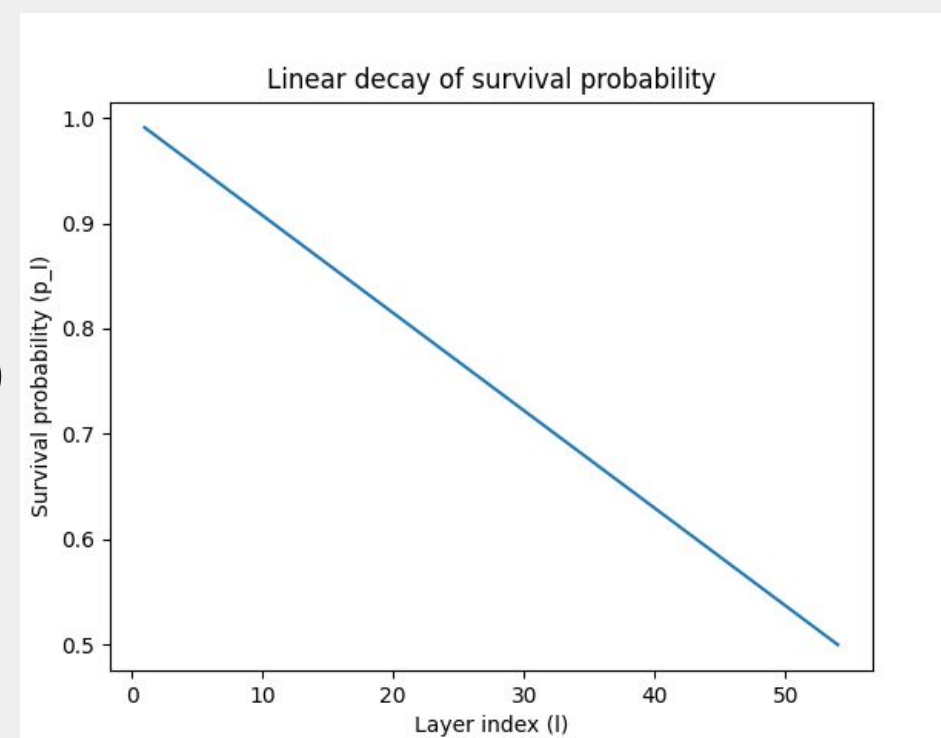


Figure 3. Linear decay of survival probability across 54 residual blocks with $p_L = 0.5$.

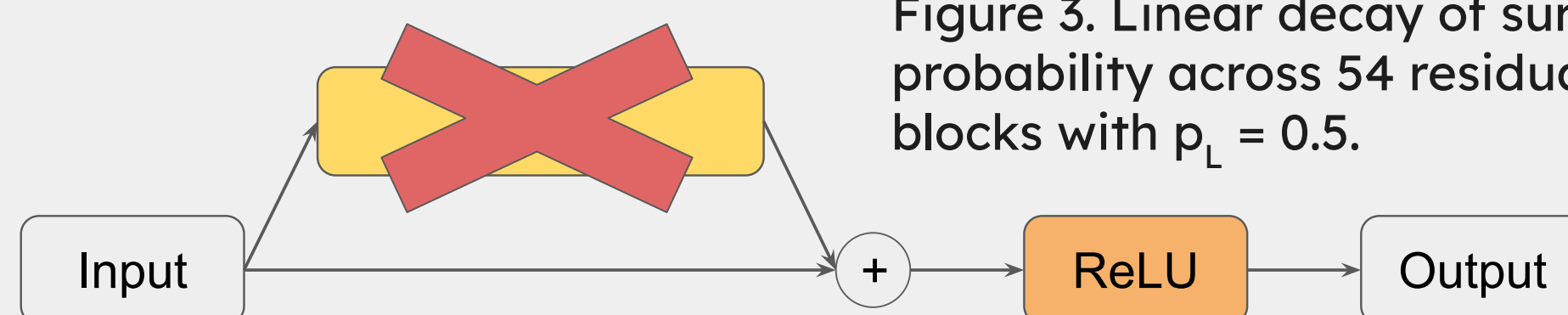


Figure 2. Residual block with dropped transformation (stochastic depth, $b_l=0$).

Re-implementation Setup

Architecture

- 110-layer ResNet
- CIFAR-10

Modifications

- 1x1 Conv → BN projection skip connections vs. avg pool + zero-pad
- Drop decision inside block's forward() vs. external gate loop

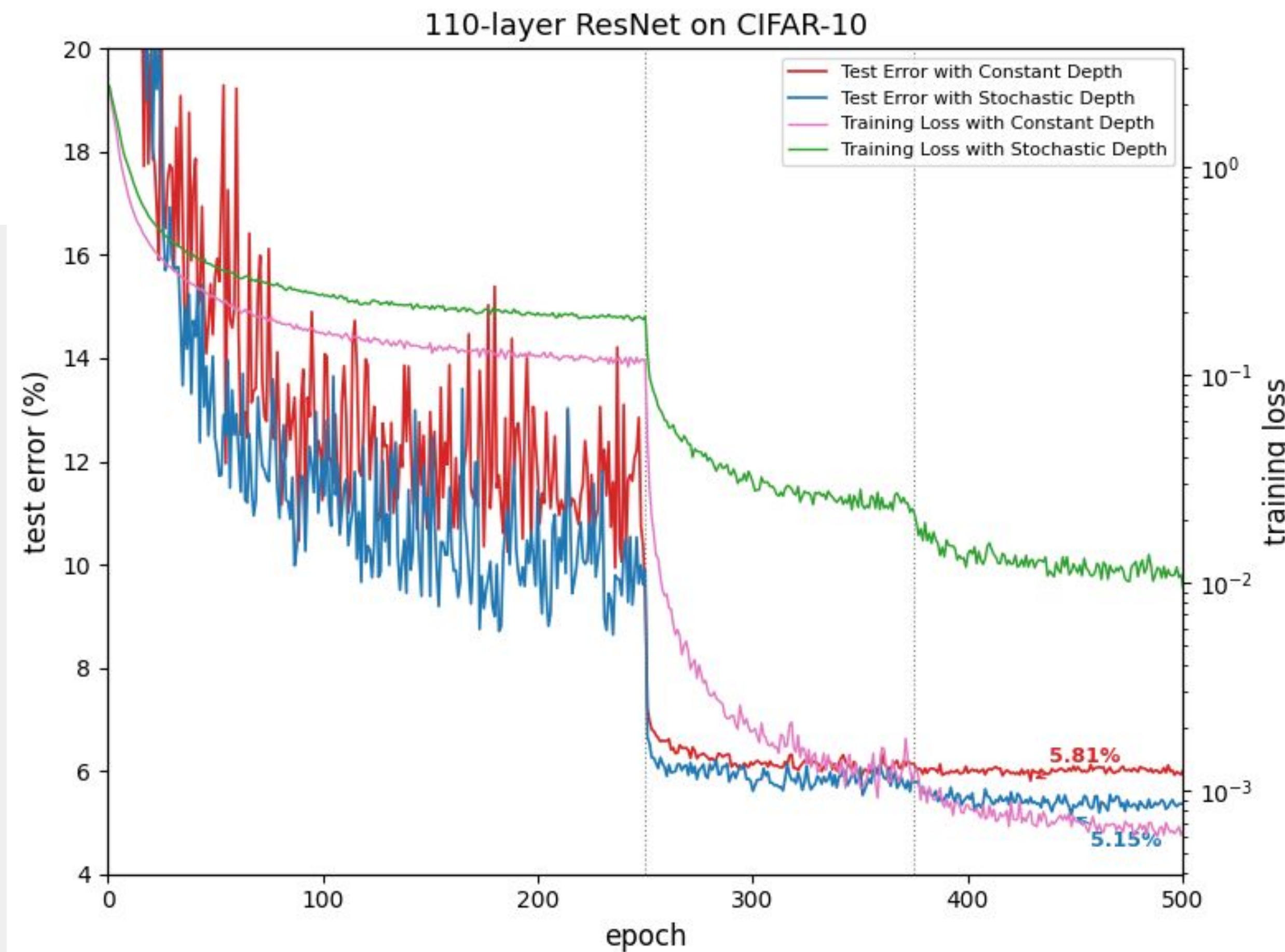


Figure 4. Test error and training loss for 110-layer ResNet on CIFAR-10. Dotted lines indicate learning rate drops.

References

- Huang, G., Sun, Y., Liu, Z., Sedra, D., and Weinberger, K.Q. — "Deep Networks with Stochastic Depth," ECCV 2016, arXiv:1603.09382.
- He, K., Zhang, X., Ren, S., and Sun, J. — "Deep Residual Learning for Image Recognition," CVPR 2016, arXiv:1512.03385.
- Claude by Anthropic

Model	Paper	Re-implementation
Constant depth	6.41%	5.81%
Stochastic depth	5.25%	5.15%
Relative Improvement ↓	18.1%	15.5%

Table 1. CIFAR-10 test error comparison (Huang et al., 2016 vs Reimplementation)

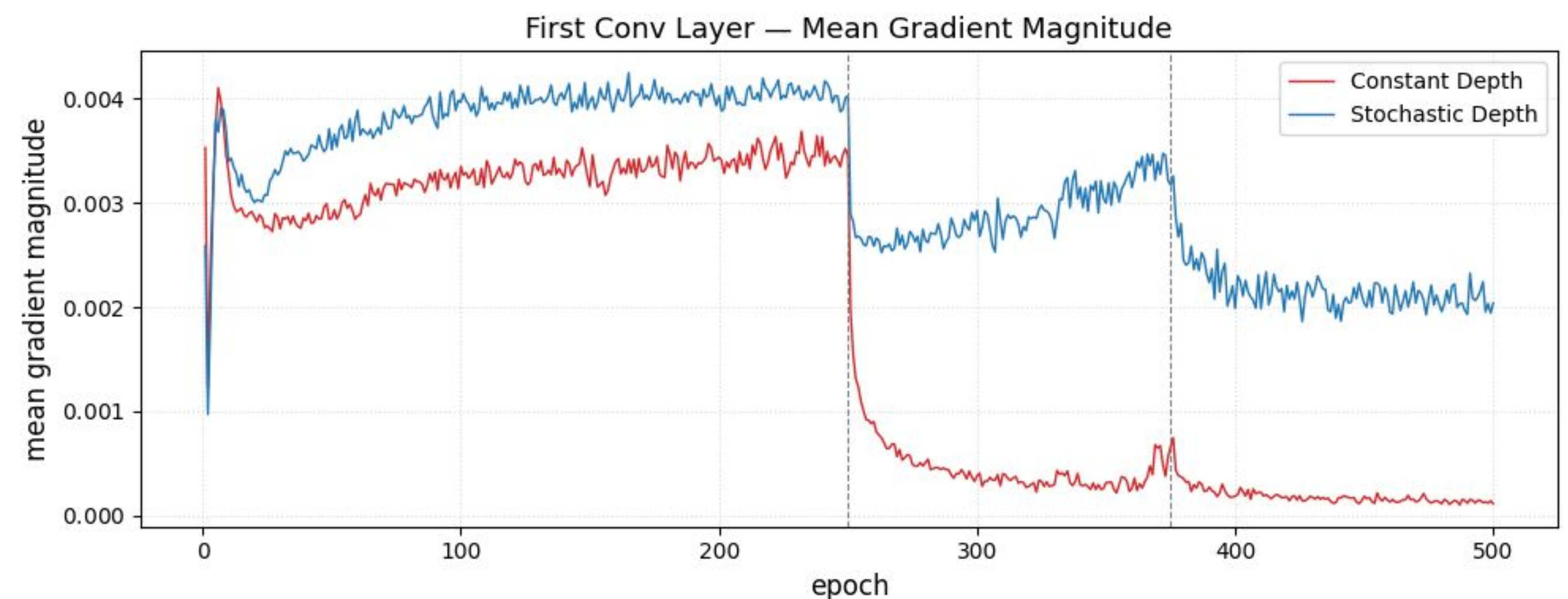


Figure 5. Mean gradient magnitude of the first convolutional layer during training. After epoch 250, constant depth gradients collapse while stochastic depth remains healthy. Dotted lines indicate learning rate drops.